[12]:
```
import pandas as pd
df = pd.read_csv("lap.csv")
print(df)
pd.get_dummies(df)
from sklearn.preprocessing import OneHotEncoder
enc = OneHotEncoder()
enc_data = enc.fit_transform(df[['Category']])
print(enc_data)
print(enc.categories_)
enc_data.toarray()
enc_df = pd.DataFrame(enc_data.toarray())
print(enc_df)
enc_df = pd.DataFrame(emc_data.toarray(), columns = ['2 in 1 Convertible', 'Gaming', 'Netbook', 'Notebook', 'Ultrabook', 'Workstation'])
print(enc_df)
df1 = df.join(enc_df)
print(df1)
```

```
    Manufacturer       Model_Name   Category  Screen_size_inches  \
0          Apple      MacBook Pro  Ultrabook                13.3
1          Apple      Macbook Air  Ultrabook                13.3
2             HP           250 G6   Notebook                15.6
3          Apple      MacBook Pro  Ultrabook                15.4
4          Apple      MacBook Pro  Ultrabook                13.3
..           ...              ...        ...                 ...
972         Dell     Alienware 17     Gaming                17.3
973      Toshiba  Tecra A40-C-1DF   Notebook                14.0
974         Asus        Rog Strix     Gaming                17.3
975           HP      Probook 450   Notebook                15.6
976       Lenovo    ThinkPad T460   Notebook                14.0

                                Screen                          CPU   RAM  \
0    IPS Panel Retina Display 2560x1600         Intel Core i5 2.3GHz   8GB
1                             1440x900         Intel Core i5 1.8GHz   8GB
2                     Full HD 1920x1080   Intel Core i5 7200U 2.5GHz   8GB
3    IPS Panel Retina Display 2880x1800         Intel Core i7 2.7GHz  16GB
```

[13]:
```
from sklearn.preprocessing import LabelEncoder
```

[13]:
```python
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['RAM'] = le.fit_transform(df['RAM'])
print(df)
print(le.classes_)
```

```
    Manufacturer       Model_Name   Category  Screen_size_inches  \
0          Apple      MacBook Pro  Ultrabook                13.3
1          Apple      Macbook Air  Ultrabook                13.3
2             HP           250 G6   Notebook                15.6
3          Apple      MacBook Pro  Ultrabook                15.4
4          Apple      MacBook Pro  Ultrabook                13.3
..           ...              ...        ...                 ...
972         Dell     Alienware 17     Gaming                17.3
973      Toshiba  Tecra A40-C-1DF   Notebook                14.0
974         Asus        Rog Strix     Gaming                17.3
975           HP      Probook 450   Notebook                15.6
976       Lenovo    ThinkPad T460   Notebook                14.0

                               Screen                          CPU  RAM  \
0    IPS Panel Retina Display 2560x1600         Intel Core i5 2.3GHz    7
1                             1440x900         Intel Core i5 1.8GHz    7
2                     Full HD 1920x1080   Intel Core i5 7200U 2.5GHz    7
3    IPS Panel Retina Display 2880x1800         Intel Core i7 2.7GHz    1
4    IPS Panel Retina Display 2560x1600         Intel Core i5 3.1GHz    7
..                                 ...                          ...  ...
972                   Full HD 1920x1080  Intel Core i7 6700HQ 2.6GHz    4
973                   Full HD 1920x1080   Intel Core i5 6200U 2.3GHz    7
974                   Full HD 1920x1080  Intel Core i7 7700HQ 2.8GHz    1
975         IPS Panel Full HD 1920x1080  Intel Core i5 7200U 2.70GHz    7
976                             1366x768   Intel Core i5 6200U 2.3GHz    5

                  Storage                          GPU Operating_System  \
0              128GB SSD  Intel Iris Plus Graphics 640            macOS
1     128GB Flash Storage        Intel HD Graphics 6000            macOS
2              256GB SSD         Intel HD Graphics 620            No OS
3              512GB SSD            AMD Radeon Pro 455            macOS
4              256GB SSD  Intel Iris Plus Graphics 650            macOS
```

```
972                   Full HD 1920x1080  Intel Core i7 6700HQ 2.6GHz    4
973                   Full HD 1920x1080   Intel Core i5 6200U 2.3GHz    7
974                   Full HD 1920x1080  Intel Core i7 7700HQ 2.8GHz    1
975         IPS Panel Full HD 1920x1080  Intel Core i5 7200U 2.70GHz    7
976                            1366x768   Intel Core i5 6200U 2.3GHz    5

                  Storage                          GPU Operating_System  \
0               128GB SSD  Intel Iris Plus Graphics 640            macOS
1     128GB Flash Storage        Intel HD Graphics 6000            macOS
2               256GB SSD         Intel HD Graphics 620            No OS
3               512GB SSD            AMD Radeon Pro 455            macOS
4               256GB SSD  Intel Iris Plus Graphics 650            macOS
..                    ...                            ...              ...
972  256GB SSD +  1TB HDD       Nvidia GeForce GTX 1070          Windows
973             256GB SSD         Intel HD Graphics 520          Windows
974  256GB SSD +  1TB HDD       Nvidia GeForce GTX 1060          Windows
975  128GB SSD +  1TB HDD          Nvidia GeForce 930MX          Windows
976          508GB Hybrid         Intel HD Graphics 520          Windows

    Operating_System_Version  Weight_kg        Price
0                        NaN       1.37  11912523.48
1                        NaN       1.34   7993374.48
2                        NaN       1.86   5112900.00
3                        NaN       1.83  22563005.40
4                        NaN       1.37  16037611.20
..                       ...        ...          ...
972                       10       4.42  24897600.00
973                       10       1.95  10492560.00
974                       10       2.73  18227710.80
975                       10       2.04   8705268.00
976                        7       1.70   8909784.00

[977 rows x 13 columns]
['12GB' '16GB' '24GB' '2GB' '32GB' '4GB' '6GB' '8GB']
```

[ ]: